PROJECT DOCUMENTATION

SAP P2P MANAGEMENT SYSTEM

Procure-to-Pay Lifecycle & Financial Integration Report

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Course: SAP DATA ANALYTICS

Project Reference: Procure-to-Pay (P2P) - Full Purchasing Cycle (SAP MM)

Live Demo : https://sap-project-p2-p-1ehl.vercel.app/dashboard

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1. Problem Statement

In modern organizations, procurement activities often involve multiple departments, vendors, approvals, goods verification, invoice validation, and payment processing. When these activities are handled manually or through disconnected systems, it becomes difficult to maintain transparency, control spending, prevent duplicate payments, and track the exact status of each purchasing document.

This project addresses that gap by creating a web-based SAP P2P simulation that demonstrates the complete purchasing lifecycle with status-based document flow, master data usage, transaction-code navigation, and financial tracking.

2. Solution Overview

The SAP P2P Management System is a web-based simulation of the Procure-to-Pay lifecycle. It models the major business stages used in SAP Materials Management and Financial Accounting, including purchase requisition creation, purchase order conversion, goods receipt confirmation, invoice processing, and final vendor payment. The system provides a single operational dashboard where users can monitor total requests, purchase orders, received goods, invoices, paid transactions, pending work, and supplier spend. It improves the learning experience by combining SAP-style business logic with a modern and easy-to-use user interface.

3. Core Features

3.1 End-to-End P2P Lifecycle

The application simulates the five major stages of the procurement lifecycle:

Document Type	Standard T-Code	Action Description	Financial Impact
Requisition (PR)	ME51N	Internal request created by a cost center.	None (Commitment only)
Purchase Order (PO)	ME21N	Legal contract issued to Vendor Master.	Encumbrance created
Goods Receipt (GR)	MIGO	Logistics confirmation of delivery.	Stock/Expense update
Invoice Receipt (IR)	MIRO	Verification of bill against receipt.	Liability (AP) recognized
Payment (FI)	F-53	Settlement via UPI/Bank/Card.	Cash outflow

Each stage updates the transaction status and document number, allowing the user to understand how procurement documents move through an ERP system.

3.2 Master Data Governance

The system uses predefined vendor, material, and cost center master data. This helps reduce manual entry errors and demonstrates how ERP systems depend on controlled master records for accurate procurement and financial processing.

3.3 Three-Way Match Concept

The project demonstrates the idea of controlled payment processing by ensuring that goods are received and invoices are verified before payment can be completed. This reflects the real-world three-way match between Purchase Order, Goods Receipt, and Invoice.

3.4 Real-Time Dashboard and Analytics

The dashboard provides procurement visibility through:

- **Procurement Throughput:** Tracks total created requests, purchase orders released, and goods received to monitor the end-to-end flow.
- **Financial Performance:** Summarizes total paid spend and paid transactions alongside a breakdown of top-performing suppliers.
- **Inventory & Backlog:** Provides visibility into pending orders and pending invoices to identify potential bottlenecks.
- **Operational Health:** Displays workflow completion percentages and operational signals for open work and invoice exposure.

4. Tech Stack

This document outlines the core technologies and architectural decisions selected for the project. The stack focuses on modern performance standards, developer efficiency, and a streamlined deployment pipeline.

Frontend: Next.js 15 (App Router Architecture)

UI Library: React.js 19

Logic: React.js Hooks for client-side state management

Styling: Tailwind CSS for responsive and professional user interface design

Persistence: Browser LocalStorage for no-database persistent simulation data

Icons: Lucide-React for clean enterprise-style iconography

Routing: Next.js App Router for page-based navigation

Deployment: Vercel for hosting and live web deployment

Data Handling: JavaScript objects and arrays for mock SAP master data and transaction records

State Storage: Custom useOrders hook for managing requisitions, orders, invoices, and payments

5. System Modules

- **Command Center (Dashboard):** Provides a high-level overview of procurement activity, workflow progress, financial summaries, and real-time supplier spend analytics.
- **Purchase Requisitions:** Enables creation of procurement requests using master material data or manual entries, capturing essential details like quantity, vendor, price, and cost center logic.
- **Purchase Orders:** Facilitates the conversion of approved requisitions into official purchase orders, complete with a document register and detailed order previews.
- **Goods Receipt (GR):** Manages the confirmation of received goods or services, advancing the document flow from the ordering phase to physical receipt.
- **Invoice Verification:** Simulates the SAP MIRO process by verifying received goods against supplier invoices to prepare the transaction for final settlement.
- **Payment Processing:** Executes vendor payments via various methods and archives completed transactions for historical tracking and spend analysis.

6. System Screenshots

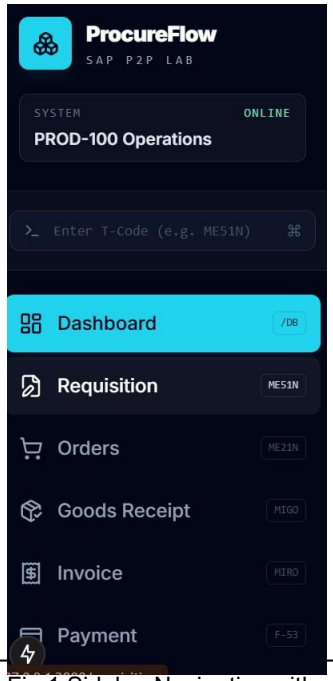

Fig.1:Sidebar Navigation with T-Code Command Interface

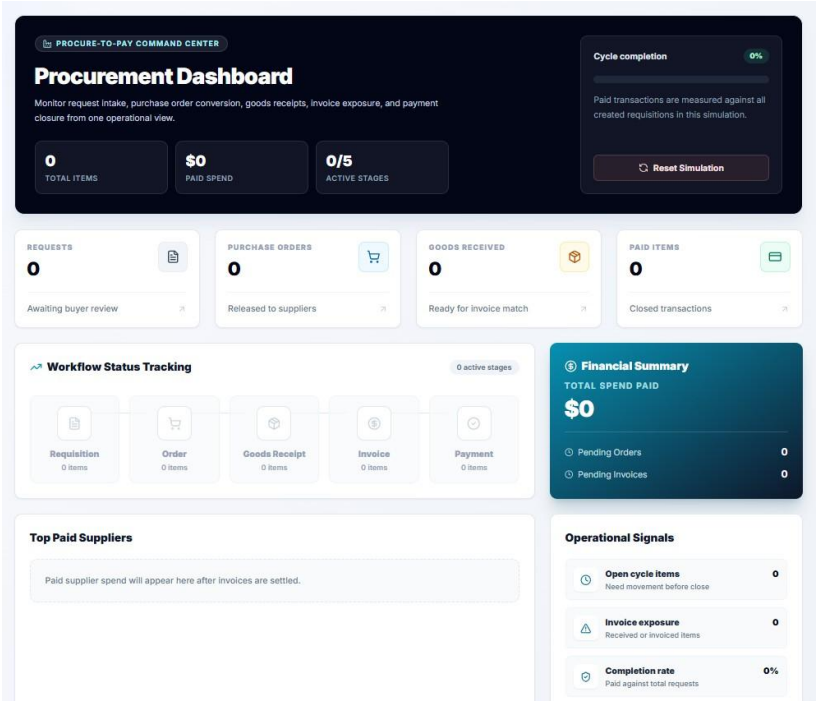

Fig 2:Procurement Dashboard with Workflow Tracking and Financial Summary

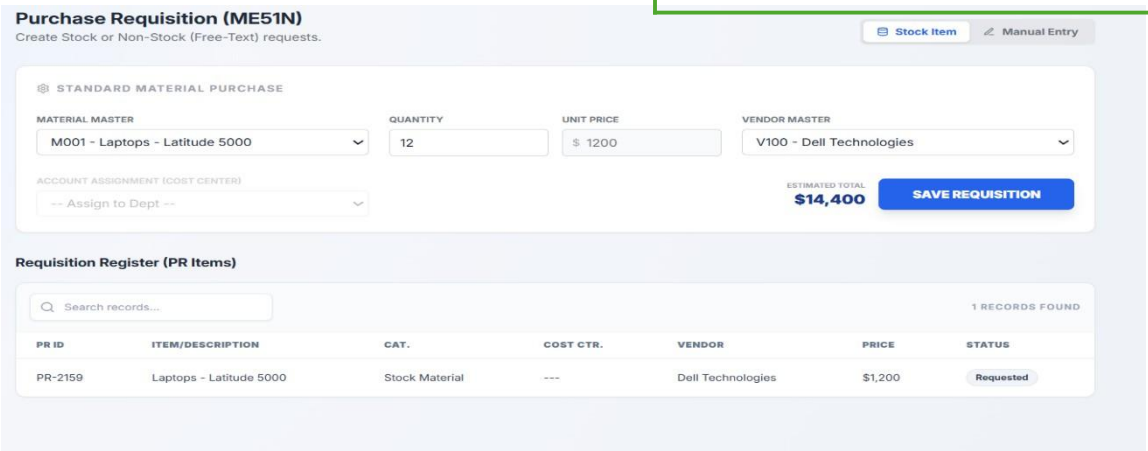

Fig 3:Purchase Requisition Interface with Material and Vendor Selection

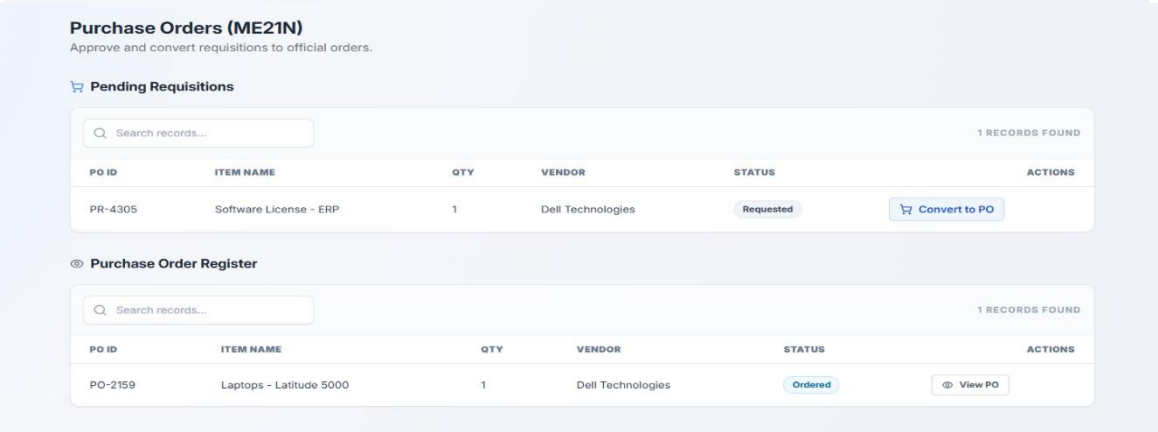

Fig 4: Purchase Order Interface with Requisition Conversion and PO Register

Goods Receipt

Log incoming shipments and verify items.

Search records...

1 RECORDS FOUND

PO ID	ITEM NAME	QTY	VENDOR	STATUS	ACTIONS
PO-2159	Laptops - Latitude 5000	1	Dell Technologies	Ordered	Confirm Receipt

Fig 5: Goods Receipt Interface with Receipt Confirmation

Invoice Verification

Match supplier invoices with goods receipts.

Search records...

1 RECORDS FOUND

GR ID	ITEM NAME	TOTAL VALUE	VENDOR	STATUS	ACTIONS
GR-2159	Laptops - Latitude 5000	$NaN	Dell Technologies	Received	Verify Invoice

Fig 6:Invoice Verification Interface with

Payments

Process final payments for verified invoices.

Pending Payments

Search records...

1 RECORDS FOUND

ID	ITEM NAME	AMOUNT	VENDOR	STATUS	ACTIONS
INV-4466	Software License - ERP	$NaN	Dell Technologies	Invoiced	Card UPI

Payment History

Search records...

1 RECORDS FOUND

ID	ITEM NAME	AMOUNT	VENDOR	METHOD	STATUS
PAY-2159	Laptops - Latitude 5000	$NaN	Dell Technologies	Card	Paid

Fig 7: Payment Interface with Pending Payments and Payment History

Purchase Order PO-5098

SAP P2P CORP

ENTERPRISE PROCUREMENT DIVISION

ORDERED

Date: 20/04/2026

Doc #: 1776701265098

123 ERP Avenue, Tech City
Digital State, 56789

SHIP TO:

Central Warehouse A1
Unit 45, Logi-Park
Sector 9, Digital Hub

VENDOR:

Amazon Web Services
Approved SAP Partner
Certified Supplier Network

DESCRIPTION	QUANTITY	PRICE	AMOUNT
Software License - ERP	1	$2,500	$2,500
		GRAND TOTAL	$2,500

1. This is a computer-generated document for SAP Simulation purposes.
2. Subject to standard ERP procurement terms and conditions.
3. Electronic verification ID: PO-5098-1776701265098

Fig 8:Purchase Order Document Preview with Vendor and Financial Details

7. Unique Points

- **Core Architecture**: Implements SAP-style T-Code navigation for rapid module access and simulates the complete Procure-to-Pay (P2P) document flow.
- **Data Management**: Utilizes master data for vendors, materials, and cost centers, supporting both stock and non-stock procurement scenarios.
- **Validation & Logic**: Enforces cost center logic for non-stock requisitions and demonstrates three-way match concepts prior to payment.
- **User Interface**: Features a modern, responsive dashboard with procurement analytics and searchable registers for document tracking.
- **System Capabilities**: Includes document previews for purchase orders and utilizes browser LocalStorage to preserve simulation data without an external database.

8. Future Improvements

- **Governance & Access**: Implement high-value approval workflows and role-based access control (RBAC) for requesters, buyers, warehouse staff, accountants, and managers.
- **Infrastructure & Security**: Migrate from LocalStorage to Supabase/PostgreSQL for multi-user persistence, add secure authentication, and enable audit logs for all document status changes.
- **Financial & Document Tools**: Add multi-currency and tax calculation support, alongside automated PDF and email generation for purchase orders.
- **Analytics & Vendor Management**: Integrate spend analysis charts and track vendor performance through ratings and delivery metrics.

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9. Conclusion

The SAP P2P Management System successfully demonstrates the complete Procure-to-Pay lifecycle in a practical and interactive way. It helps users understand how procurement documents are created, converted, verified, and financially closed within an ERP-style workflow.

By combining SAP-inspired transaction codes, master data governance, document flow, dashboard analytics, and financial controls, the project provides a strong educational simulation for learning SAP MM and FI integration concepts. The system can be further improved with approvals, database storage, role-based access, and advanced reporting to make it closer to a real enterprise procurement solution.